Ex: $A = \begin{bmatrix} 3 & 1 \\ 3 & -1 \end{bmatrix}$ for eigenvalues $|A - \lambda I| = 0$

solve for λ: λ_1, λ_2

to find the eigenvector corresponding to λ_1

$(A - \lambda_1 I)V = 0$ solve for V: V_1

so $\lambda_1 V_1$ is an eigenvalue eigenvector pair

$$\begin{vmatrix} 3-\lambda & 1 \\ 3 & -1-\lambda \end{vmatrix} = 0 = (3-\lambda)(-1-\lambda) - 3 = \lambda^2 - 2\lambda - 6$$

$$\lambda = \frac{2 \pm \sqrt{4 - 4(-6)}}{2} = 1 \pm \sqrt{7}$$

$-\sqrt{}$ $\quad V_1$

$$\begin{bmatrix} 2+\sqrt{7} & 1 \\ 3 & -2+\sqrt{7} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \vec{0} \qquad \begin{aligned} (2+\sqrt{7})x_1 + y_1 &= 0 \\ 3x_1 + (-2+\sqrt{7})y_1 &= 0 \end{aligned} \qquad y_1 = (-2-\sqrt{7})x_1$$

$$\hookrightarrow V_1 = \begin{bmatrix} 1 \\ -2-\sqrt{7} \end{bmatrix}$$

$+\sqrt{}$ $\quad V_2$ $\quad$ and

$$\begin{bmatrix} 2-\sqrt{7} & 1 \\ 3 & -2-\sqrt{7} \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \vec{0} \qquad V_2 = \begin{bmatrix} 1 \\ -2+\sqrt{7} \end{bmatrix}$$

or

$3x_1 + (-2+\sqrt{7})(-2-\sqrt{7})x_1 = 0 \quad 3x - 3x = 0 \quad x(0) = 0$ so x is free

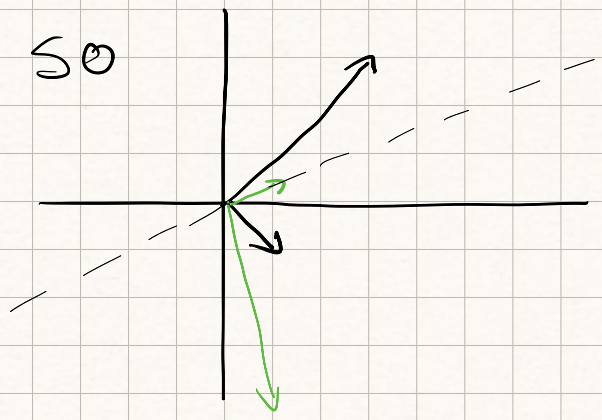
so for both λ's we get $\begin{bmatrix} x \\ (-3-\lambda)x \end{bmatrix}$

note: if A has eigenvalue λ and eigenvector v then kv is an eigenvector corresponding to λ for $k \neq 0$

$$Av = \lambda v \Rightarrow A(kv) = \lambda(kv)$$

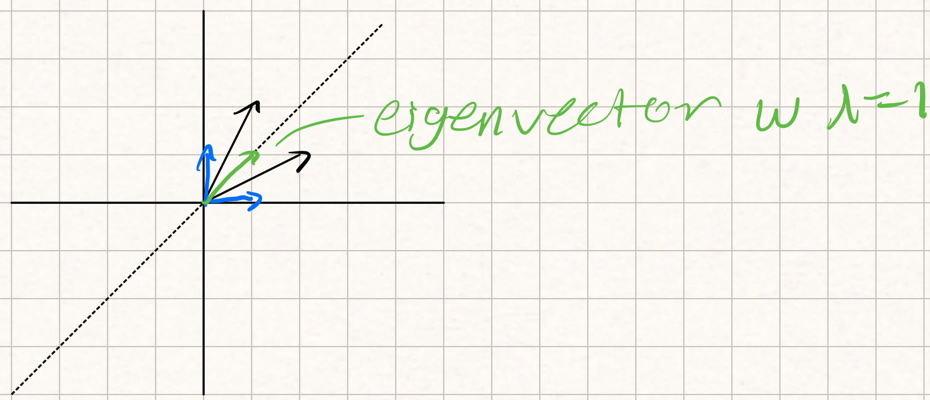
$$= x \begin{bmatrix} 1 \\ (-3+\lambda) \end{bmatrix}$$

so



in eigenvector basis $\begin{bmatrix} 1+\sqrt{7} & 0 \\ 0 & 1-\sqrt{7} \end{bmatrix}$

in $\mathbb{R}^2$ pick a line through $(0,0)$



reflection over this line is a linear transformation, what is the matrix

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$